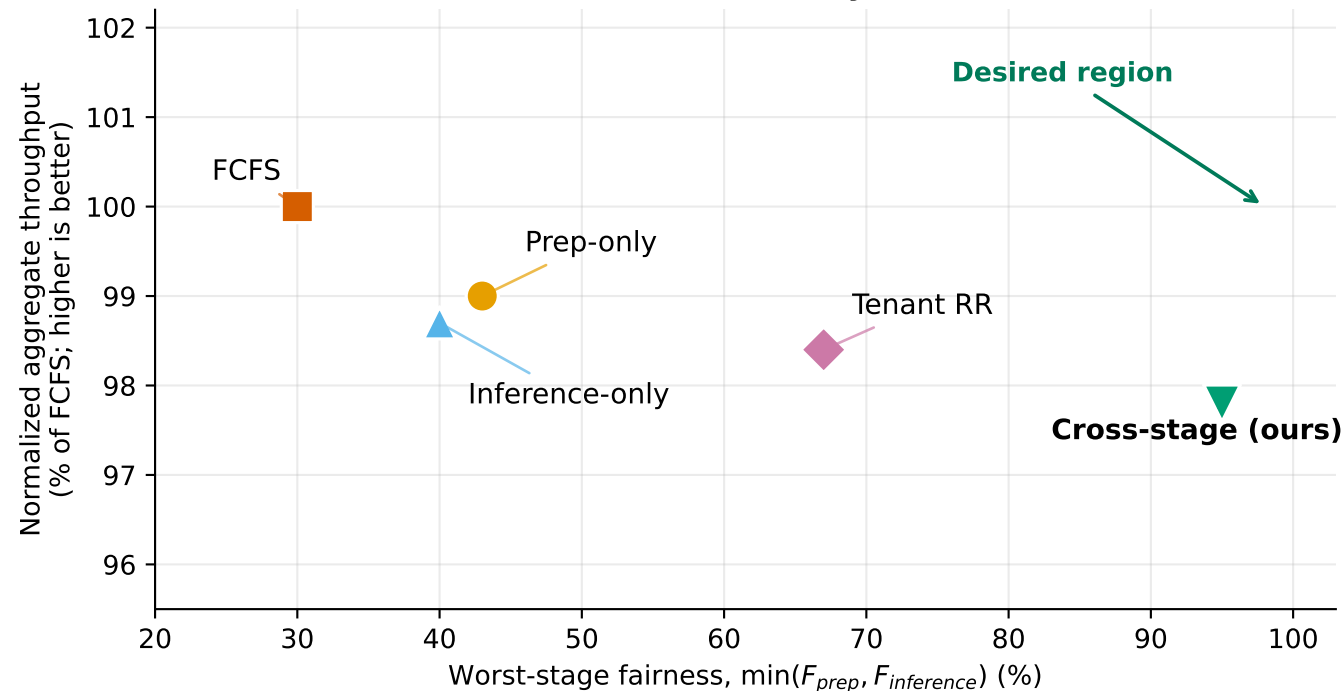


MOCKUP 9 — Fairness, Efficiency, and Work Conservation

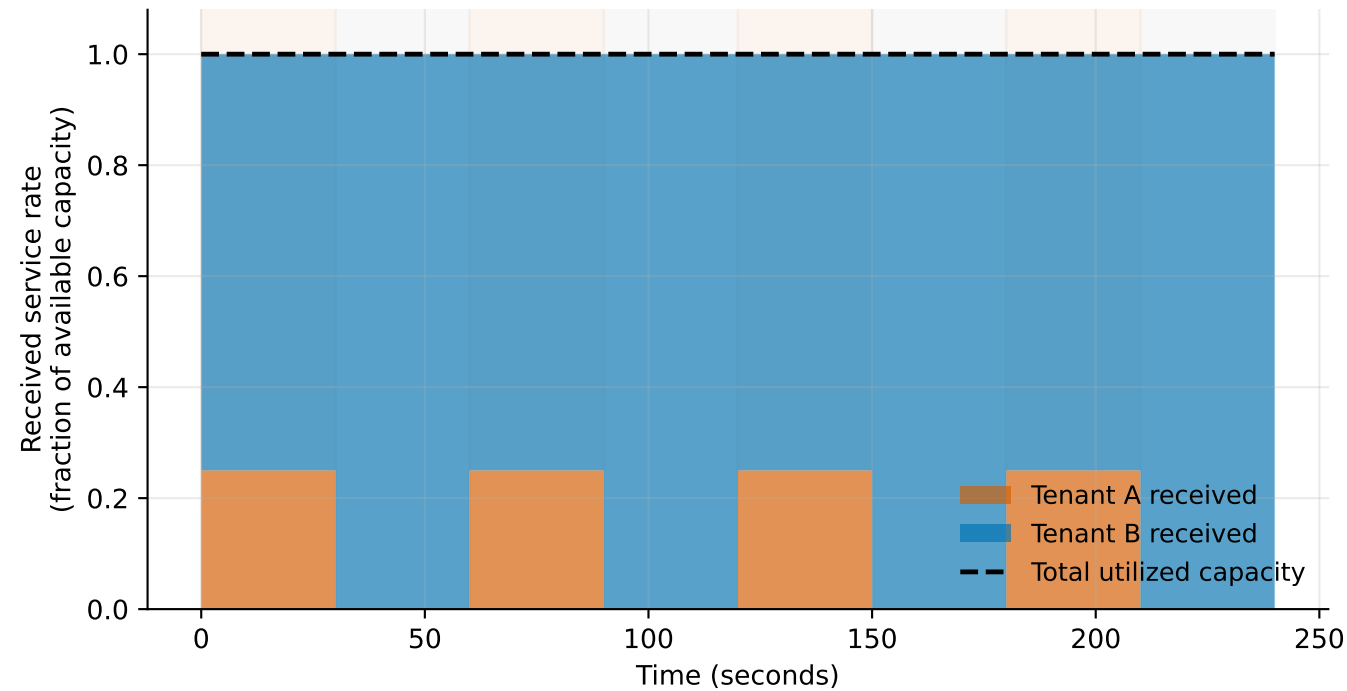
Cross-stage fairness targets tenant isolation with little throughput cost

Pareto workload: equal-rate, backlogged tenants with 32/128-frame videos and short/long outputs. ON/OFF check: A demands 0.25C while ON; B is always backlogged.

(a) Fairness-efficiency tradeoff



(b) Work conservation under ON/OFF demand



Expected conclusion: cross-stage approaches the upper-right fairness/throughput region and reallocates unused service instead of idling capacity.